



Design of a 2.4GHz-2.5GHz Low Noise Amplifier Using 90nm CMOS Technology

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Academic Integrity: The following design and summary report represent my own original work. I acknowledge that actual designs must be unique. I acknowledge that absolutely no sharing of schematics, subcircuits, designs, plots, write-ups, or reports will be accepted. I have not made any use of prior-semester reports or prior-semester designs. I acknowledge that students in violation of this policy will be reported to the NC State Office of Student Conduct, with repercussions including placement on academic probation and grade penalties including but not limited to receiving zero credit for the design project.

Abstract

This report presents the design and analysis of a Low-Noise Amplifier using 90 nm technology in Cadence Virtuoso, operating at a frequency range of 2.4-2.5GHz. The circuit includes a cascode structure comprised of inductors and a capacitive load for impedance matching, resistive choke, and an input capacitive filter. A multi-level hierarchy is used to model fixed off-chip parasitics which demonstrate the degenerative effects on an ideal design and individual pi-models for modelling inductors. The circuit nominally meets 8 out of 9 specifications, achieving low power consumption, noise figure and a high stability across the bandwidth. Due to certain tradeoffs discussed later in the report, the gain specification was dropped in favor of others.

Specification Table

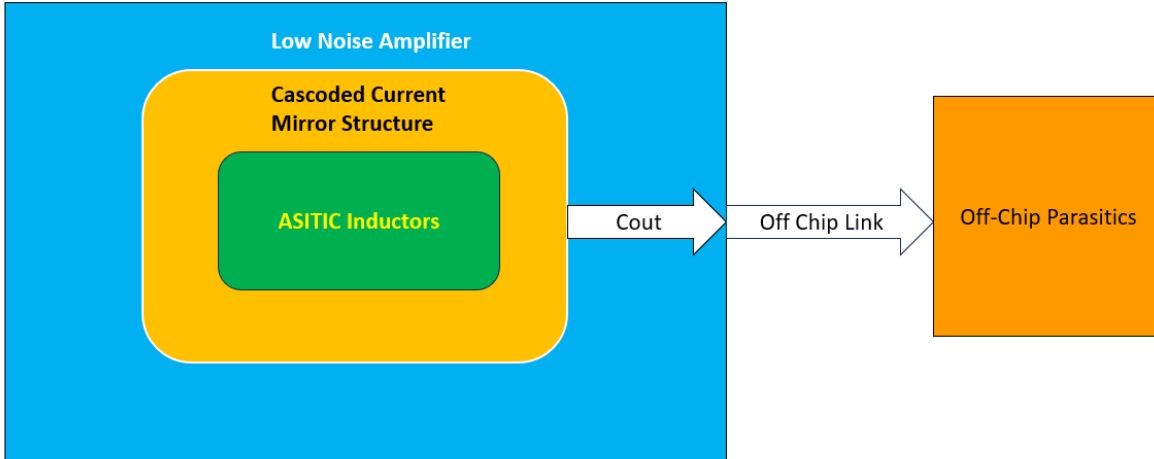
Metric	Unit	Specification	Design Value	Ref. Figure
Center Frequency	GHz	2.4 – 2.5	2.45	N/A
Noise Figure	dB	< 2.2	2.16	8
Power Consumption	mW	< 15	14.36	14
$ S_{21} $	dB	16 at center frequency	13.58	6
$ S_{11} $	dB	< -12.5 across the bandwidth	-20.06	9
$ S_{22} $	dB	< -12.5 across the bandwidth	-20.02	9
IIP3	dB	> -8	-6.26	13
iP1dB	dBm	> -18	-17.097	12
Stability	Unconditional 10MHz – 100 GHz		Stable	11

Individual Contributions:

During the time of this project, equal work has been put in by both team members in terms of circuit designing, simulation, testing and documentation.

Design Overview:

Our design comprises of a cascode current mirror based LNA structure, with inductors at the gate, drain and the source of the output leg, followed by an output capacitor prior to going off the chip. The design process was done in three major steps as described below.



System Block Diagram

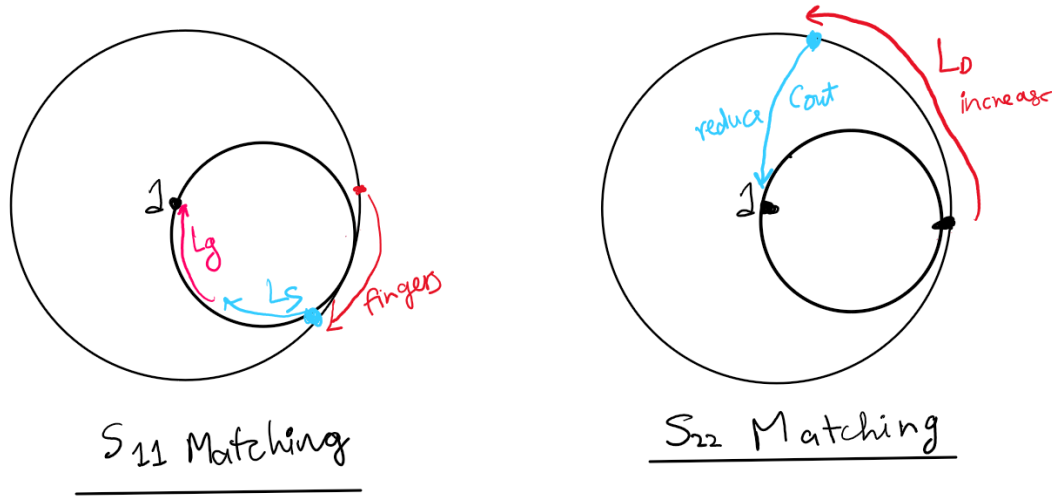
Ideal Inductor Model:

In this approach we started by having a model without any of the inductors, having a R_{big} and C_{big} at the lower gate rail with the input voltage V_s . This was done to decide a suitable operating point that gives us the match between S_{11} and G_{min} , implying that we need to find an operating point where we can get the maximum gain and the minimum noise figure. The reference current I_{ref} followed by the number of fingers at the output were swept to get this point. This was followed by a sweep of the length to get a better matching of the parameters. Meanwhile, we kept a R_{big} and C_{big} values very large, along with a C_{out} of 1F. Upon getting the optimum operating point with the appropriate dimensions, after testing several of them, and getting a fixed sizing, we inserted inductors at the respective ports. Firstly, the L_{source} values were swept, keeping the other inductors out of the pictures, followed by L_{gate} to get an optimum input match. Furthermore, the values of L_{drain} were also swept along with C_{out} to get an optimum output match. However, due to the lack of a real Q-factor in the model, we were not able to get an optimal match of both the S_{11} and S_{22} . Meanwhile, the power consumption was monitored and kept at an acceptable level. We had to intermittently change the operating point to optimize the design. Overall, all the specifications besides the gain were met across the bandwidth. G_{min} and S_{11} match were parallelly checked on a smith chart to ensure that a low noise figure was kept.

IndQ Model:

In the ideal inductor case, S_{11} and S_{22} , which are the primary means of ensuring desired operation were further matched by varying the Q-factors of all the inductors in the range of 1 to 15. The range was estimated based on the ability to create practical inductors with realistic Q-

factors in ASITIC. For our design, $Q_g > Q_d > Q_s$ was found to be a general trend. Furthermore, the process followed for matching the two in terms of component value variations for the entire design is described in the figure below.



Matching S11 and S22

The G_{min} primarily depends on the number of fingers and the L_{gate} is used to match S11 and G_{min} . To manage the power consumption, the current was varied with all components in place, and the input match was checked with the NF. For the NF, increasing the current and number of fingers decreases it, but it would consume too much power; hence, a sweet spot needs to be found out for the same. However, merely varying the Q-factors of the inductors was not sufficient, as they were up to almost 20, which was not practically feasible to achieve in the given constraints. So, the values of the inductors, output capacitor and were varied again in the same pattern as shown in the figure above and S11/22 were brought close to the 60Ω circle, meanwhile, keeping all Q-factors at a constant value between 5 to 8. They were then appropriately varied to find the optimal matching. The gain increased with the current and fingers, but it made the LNA design impractical due to a high-power consumption and loss of linearity. Refer to Figure A6 for our ideal inductances and Q factors.

ASITIC Inductor Model:

The design for the LNA requires source, gate, and drain on-chip spiral inductors. The development of these inductors was performed with the ASITIC tool. This tool solves self and mutual inductance, as well as parasitic capacitance to the grounded substrate and between turns, to produce an accurate asymmetric pi-model for the inductors. Most times, a generated ASITIC inductor deviates significantly from the ideal Q factor model, requiring several design iterations. This deviation is partially driven by shunt capacitance and lowering the resistance values within

the pi-model. The inductance and Q factor values that were collected from the simulations, using ideal inductors, served as the medium for determining whether to go over or under that inductance to achieve better impedance matching on the input and output.

We utilized the M6 metal, having the least resistance and greatest Q factor, as an entry point and the M5 layer for the exit point, as it was the furthest from the substrate. The origin was kept at the center and except for L_{source} the internal diameter D_{in} was varied. For all inductors, a unique design approach was used to reduce the resistive component and increase the Q-factor to an optimal value. While the process was tedious, intuitively varying design variables in small increments based on the approximations derived from theoretical approach of the greenhouse paper and the spiral inductor calculator made it easier to get a best estimate close to the ideal values achieved through the previous stages. The pi command makes more assumptions and gives a larger estimate than the actual value, and hence, the pix command has been used after pi during these simulations.

It was very important to make sure that all pi-models had very minimal resistance, because it would then be able to almost replicate the same behavior as an ideal inductor. Methods that were used to decrease resistance and increase Q factor were by increasing the width or having a high ratio between the width and the spacing. However, having a high metal width would make the Q factor more vulnerable to any changes in frequency. For us, although the spacing and number of turns are different for all three ports, we kept those two parameters constant when running tests for each individual port. The number of turns served as the final adjustor, because of the direct relationship it has with the inductance. We made minor changes with the spacing, because of the drastic inverse relationship that parameter has with the inductance. It was important to maintain sufficient space, to avoid disturbing the inductor's electrical characteristics. For us, the primary catalyst for finding the right inductance was by adjusting the hole diameter. For the source and the drain inductors, we had to use the outer diameter, as for such small inductance values, the resistance values are small as well, leading the resistance to go into negative whenever the diameter was increased beyond a certain limit. It's very important to know which parameters to change to find your ideal inductance in ASITIC, because even the smallest change in a single parameter could lead to a drastic change in the inductance and resistance of the pi-model.

Design Schematics:

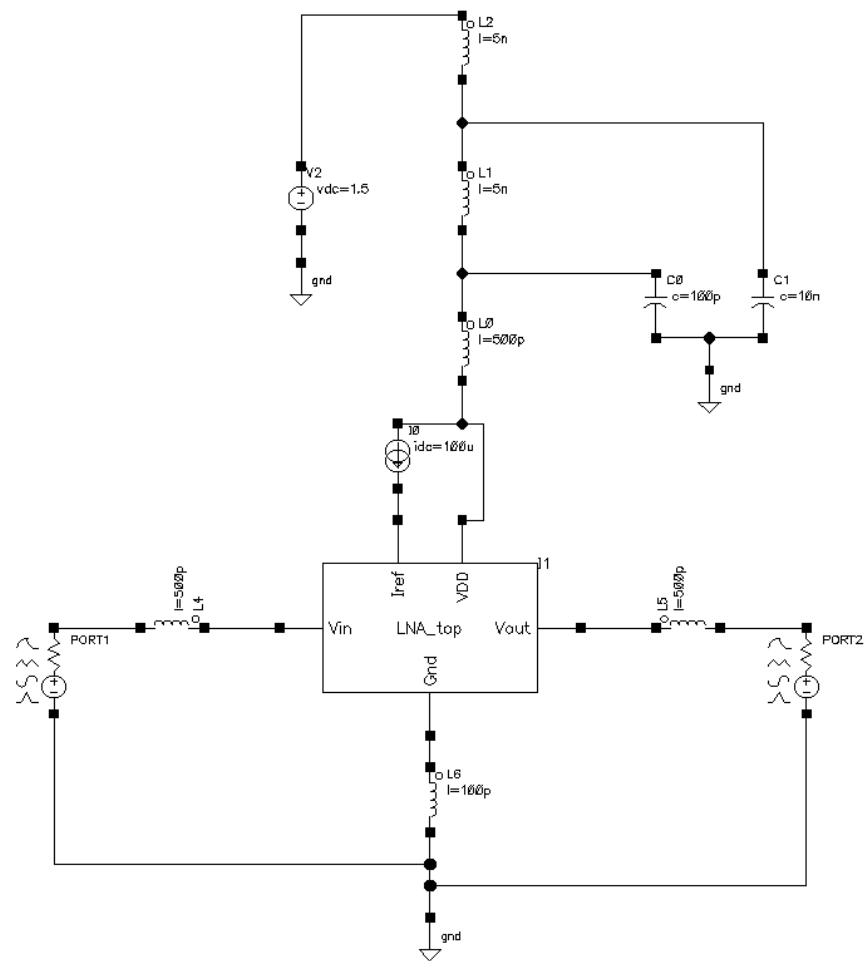


Figure 1: LNA Testbench

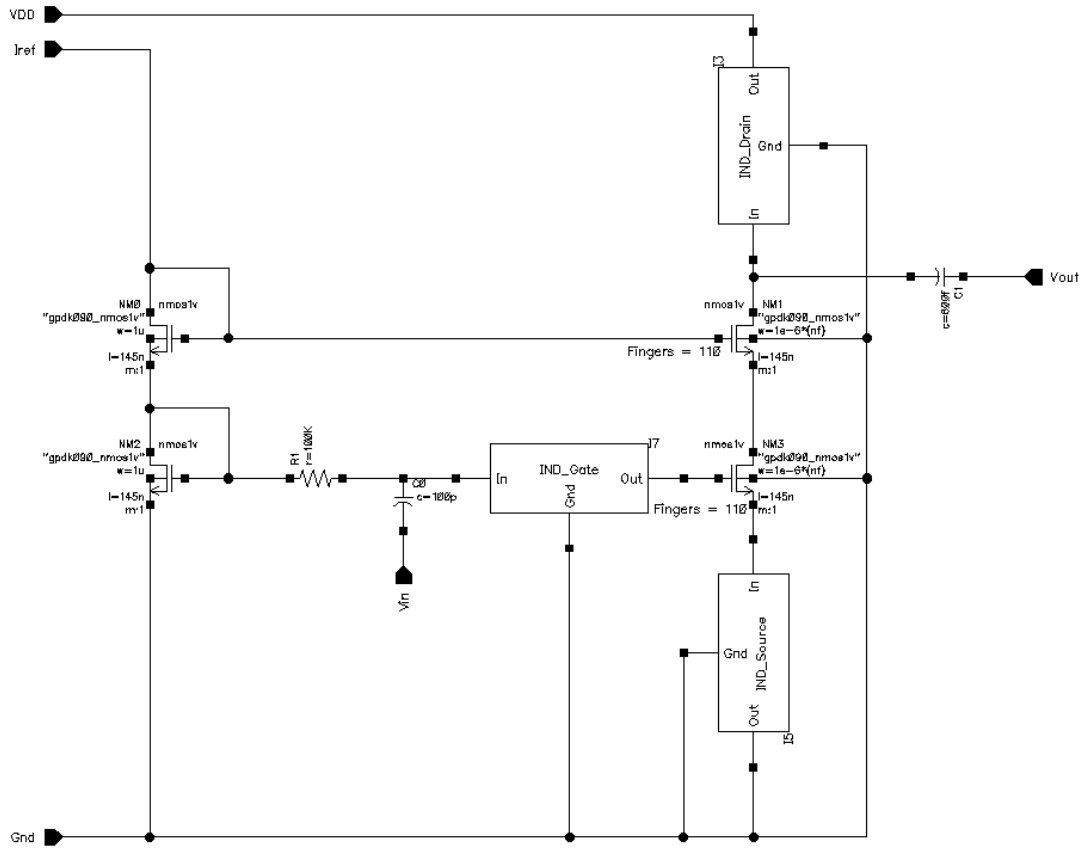
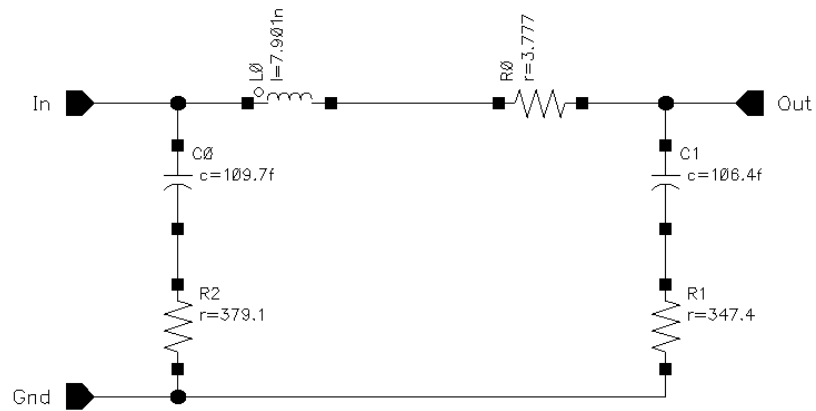


Figure 2: Top-level LNA Circuit



Inner hole = 250 um
 Width = 20 um
 Spacing = 4 um
 Turns = 3.5
 Top metal = m6
 Bottom metal = m5
 Q = 6.865, 7.277, 12.21

Figure 3: ASITIC Gate Inductor

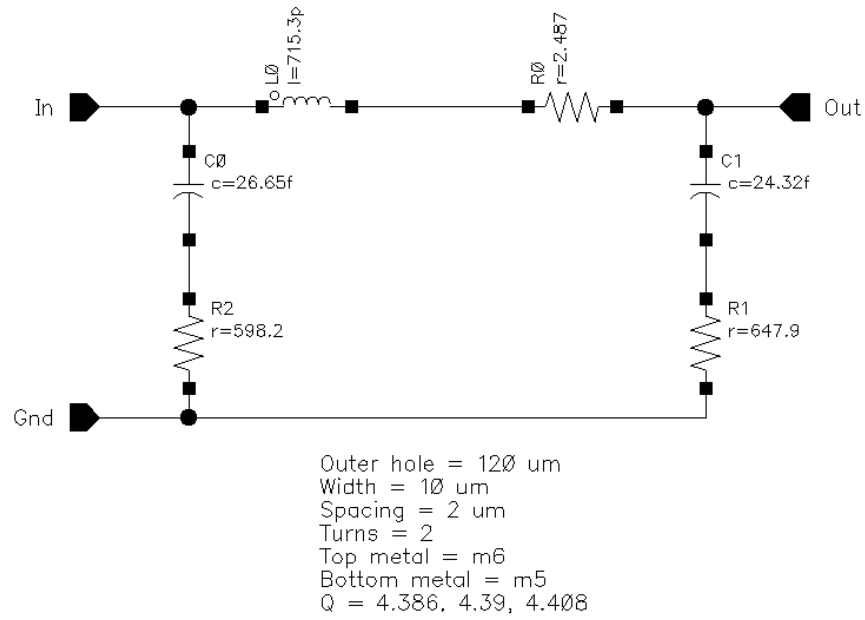


Figure 4: ASITIC Source Inductor

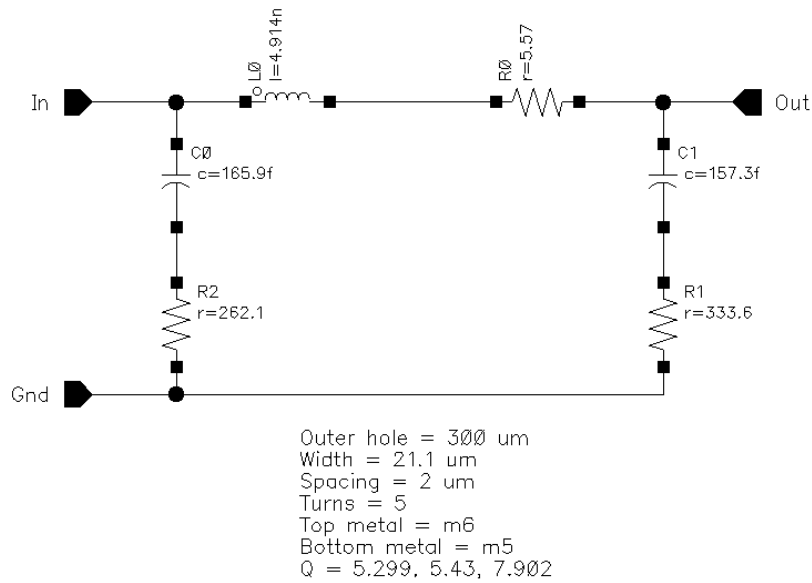


Figure 5: ASITIC Drain Inductor

Simulation Plots

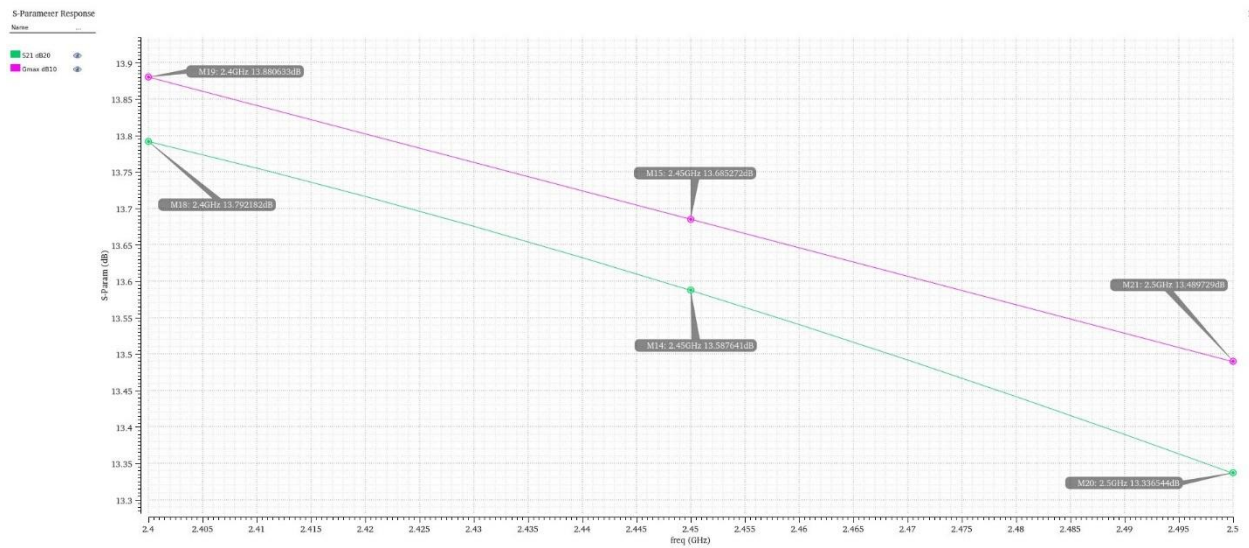


Figure 6: S21(Green) and G_{max}(Pink) in dB on Frequency

From the plot above, the gain at mid band frequency is 13.58 dB. The gain at 2.4 GHz is 13.79 dB and the gain at 2.5 GHz is 13.33 dB. The plot shows somewhat of a direct relationship between the gain and the frequency.

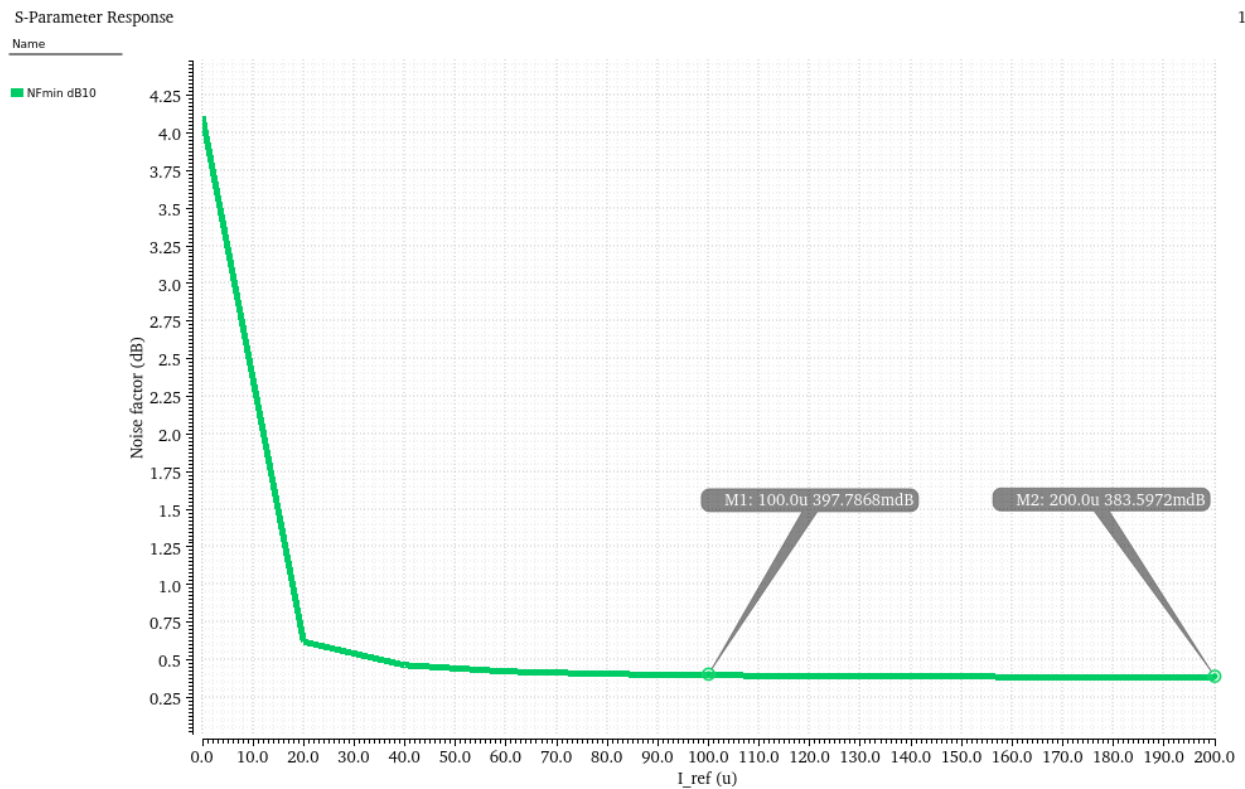


Figure 7: NF_{min} on Bias Current

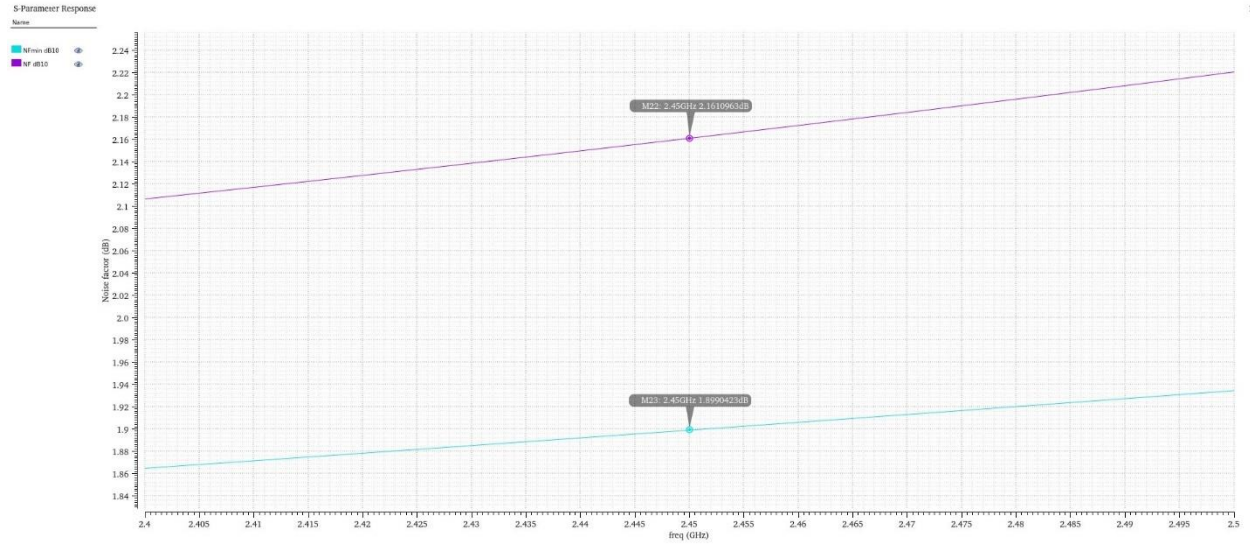


Figure 8: Noise Figure (Purple) and NF_{min} (Blue) in dB on Frequency
The noise figure is consistently below 2.2 dB for a majority of the bandwidth except from 2.48 to 2.5 GHz range but is perfectly matched at the center frequency.

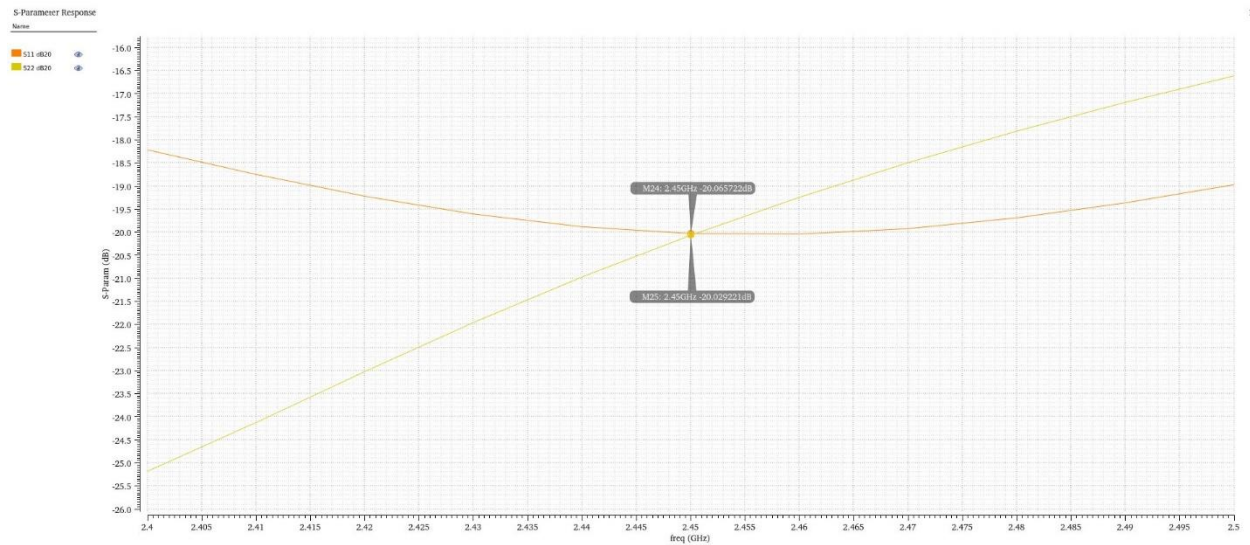


Figure 9: $|S_{11}|$ (Orange) and $|S_{22}|$ (yellow) in dB on Frequency
 S_{11} and S_{22} match perfectly at the center frequency and meet the specs across the bandwidth.

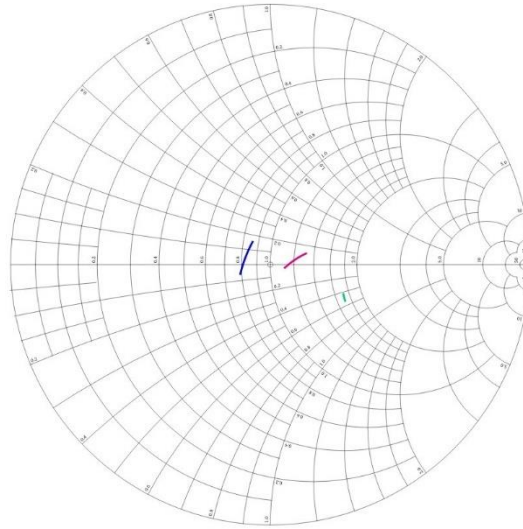


Figure 10: Impedance Smith Chart of S11(Blue), S22(Red), and Gmin(Green) over the 2.4-2.5 GHz Frequency Range

S11 and S22 are in the vicinity of the 60Ω circle but meet the requirements for the specs on the rectangular plot. $G_{\min}$ being far off from the center indicates why the NF is just matching the given specifications as matching it exactly would cause the other parameters to shift.

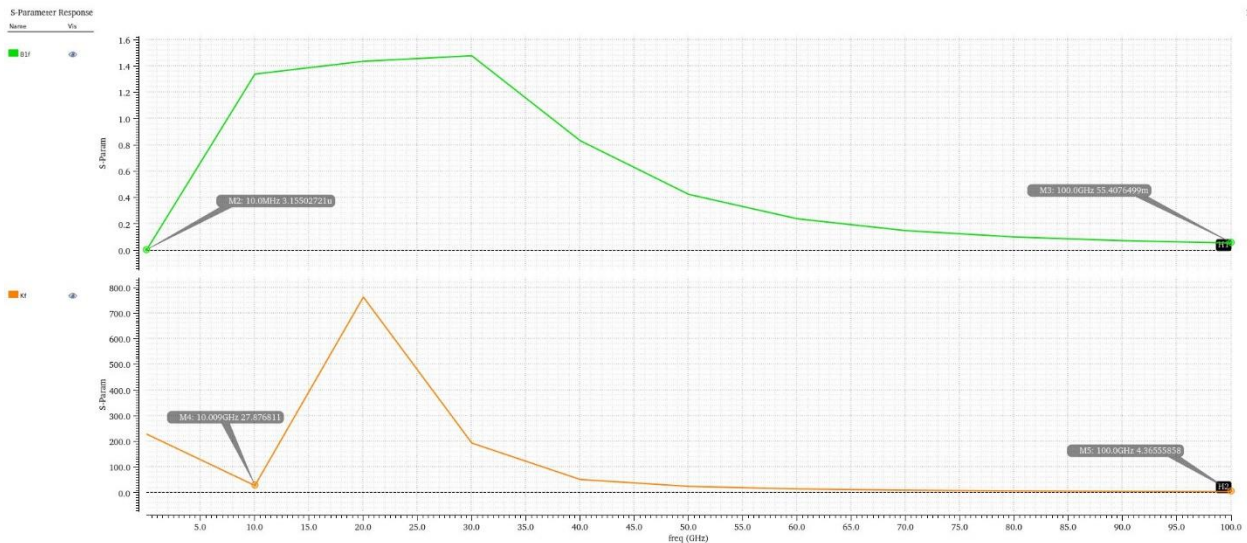


Figure 11: Stability Metrics Kf(Orange) and B1f(Green) on Frequency

For the plot above Kf is above 1 and B1f is above 0 across the bandwidth. Therefore, the design passes stability requirements.

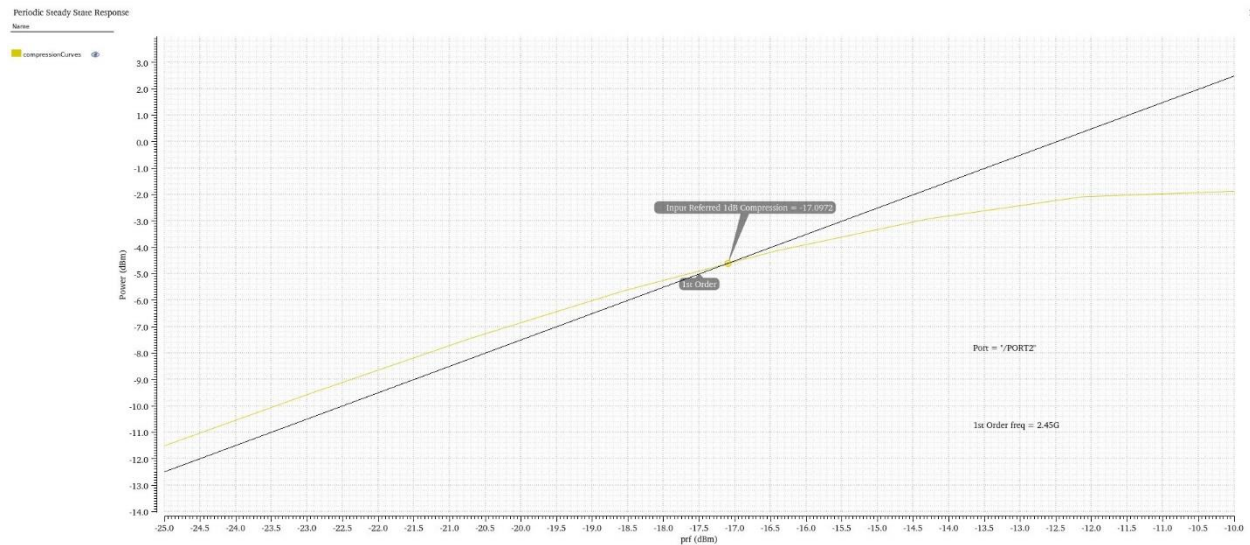


Figure 12: Input-referred P1dB in dBm (yellow marker)

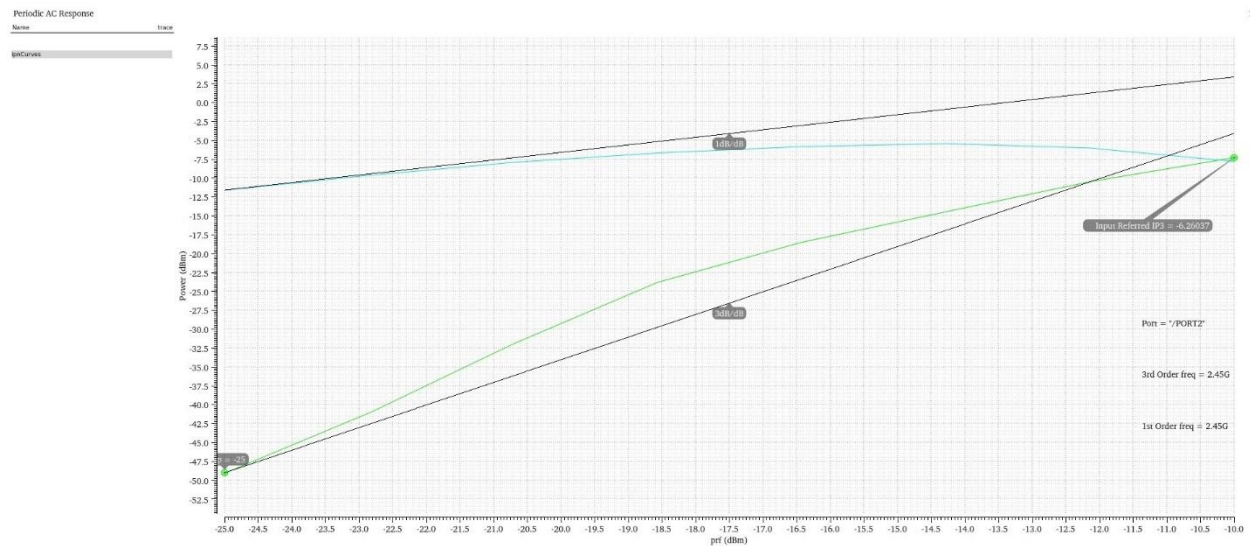


Figure 13: Input-referred IP3 in dBm (green marker)

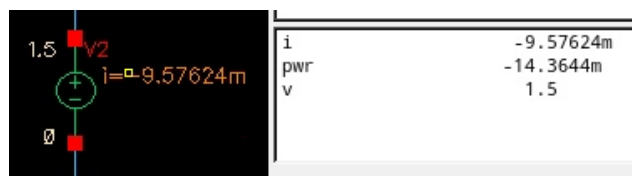


Figure 14: Total Power Consumption

Discussion

The striking detail to point out in this design is that the gain is slightly below what's required. While attempting to achieve other metrics, it was often noted that we needed to sacrifice our gain to meet the rest of the metrics in this project, hence why our gain is the only failing metric in our design. When conducting numerous tests to achieve a perfect impedance match across the input and output, there were other parameters that would also be affected, based on the variables that were being changed. For example, when wanting to decrease the noise figure, the most noted way was to increase the reference current. While increasing the current drops the noise figure, it would increase the power consumption. The same relationship can also be said about changing the number of fingers on the output side of the LNA. The alternative method was to tinker with our source inductance model and bring down in inductance in small discrete steps. It was important to bring the source inductance down in small steps, otherwise a huge decrease would result in a failing IIP3 and iP1dB.

In terms of input and output impedance matching, it was very important to consider the relationship between numerous variables such as the inductances, the number of fingers, the Q factor, and the output capacitance. For the ideal case, we could not achieve a perfect match due to the lack of a finite Q-factor; this issue was then sorted with IndQ inductors, however, it took more iterations of the Q's as well as the inductances, particularly in the gate and drain to achieve a perfect match. In all three cases, it was observed that when either S11 or S22 were perfectly matched, the other would worsen. Hence, while using real inductor models, we needed to find a particular point where these values match the metrics on a rectangular plot, despite not exactly being at the center of the smith chart.

Throughout the project, we were able to design an LNA that met stability requirements throughout all three cases of inductors only partially affecting the IIP3 and iP1dB, but still passing. For the initial iterations of the values of G_{max} and NF_{min} , if these values are far from the maximum permissible metrics allowed, then it can be said that the final metrics after the impedance matching with the ASITIC inductors will be accurate and within the specified limits. Our NF_{min} was about 1.9 dB at the center frequency, which is close to the maximum allowed value.

Conclusion

In this project an LNA for 2.4GHz to 2.5GHz range was developed using 90nm gpdK on Cadence Virtuoso, meeting 8 out of 9 specifications over the specified range and made the design as practical as possible. This included numerous iterations of various parameters, developing a realistic ASITIC-based schematic from an ideal model. During this procedure, we observed major tradeoffs such as the gain with the noise, S11 with S22 or power with gain and

understood the impact of parametric variations on them as well as internally balanced all the parameters to achieve the desired results. However, in favor of low power dissipation, good NF and linearity across a wide range, the gain had to be much lower than what was expected. At the end, variation in supply voltage, current, dimensions and finally the impedance matching can further aid us to achieve all the specifications across the entire bandwidth.

Appendix A: Schematics with annotated DC operating points and node voltages

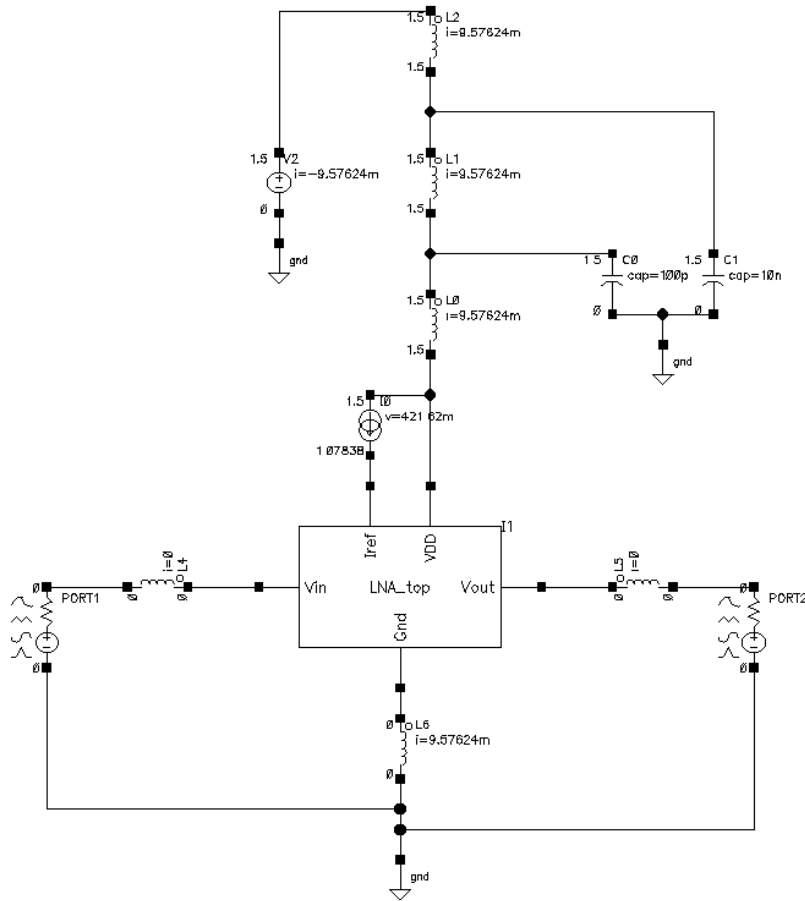


Figure A1: Annotated LNA Testbench

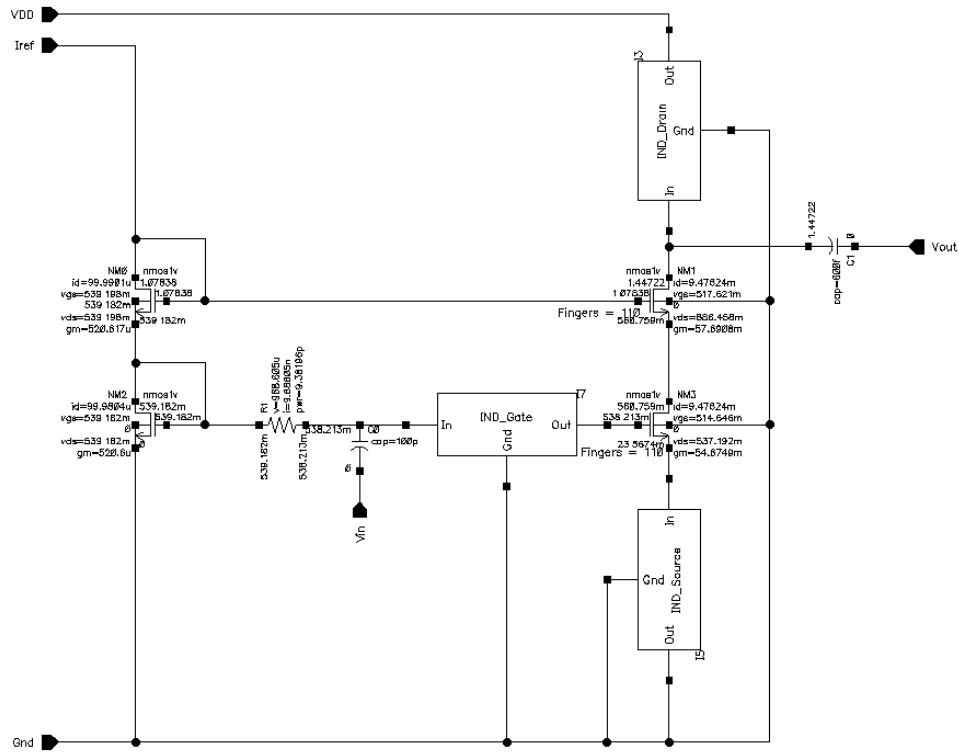


Figure A2: Annotated Top-level LNA Circuit

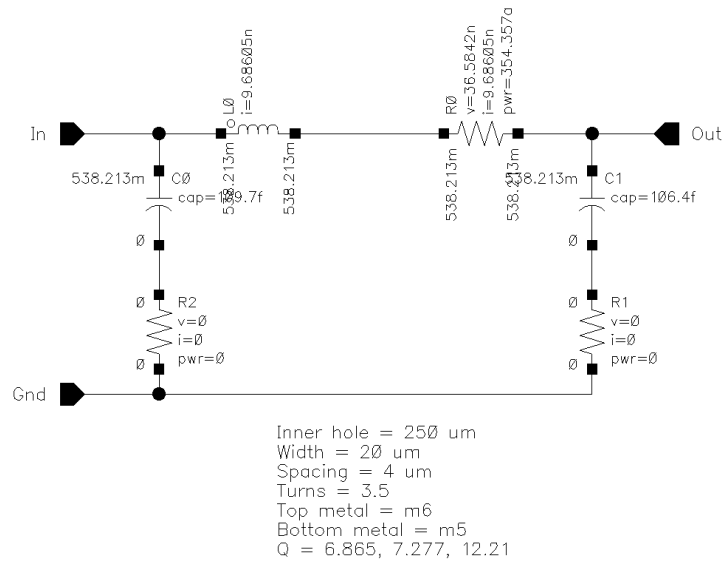


Figure A3: Annotated ASITIC Gate Inductor

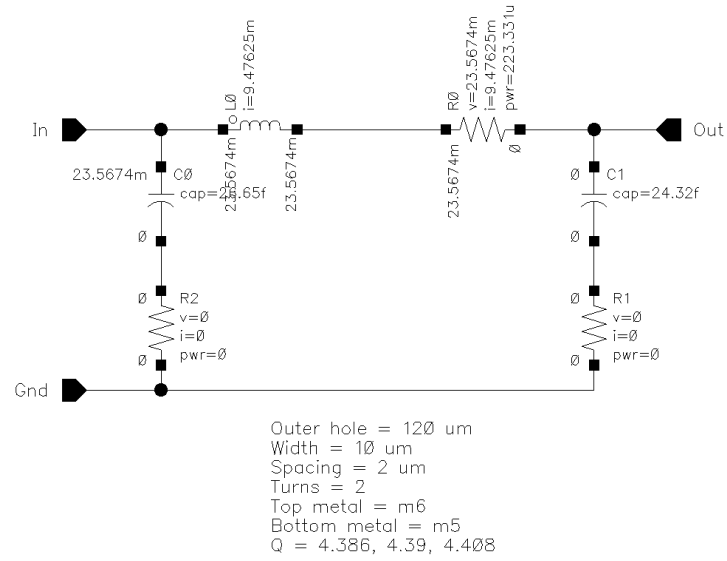


Figure A4: Annotated ASITIC Source Inductor

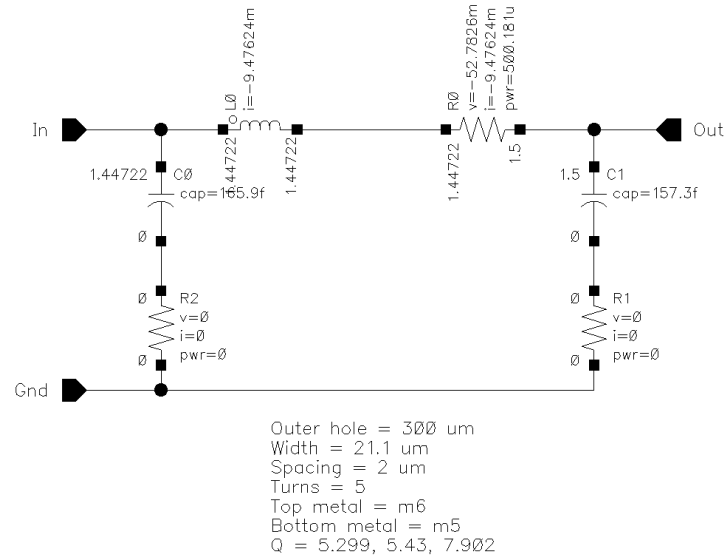


Figure A5: Annotated ASITIC Drain Inductor

Port	Inductance	Q Factor
Gate	11 nH	10
Source	1.1 nH	2
Drain	5 nH	8

Figure A6: Ideal Inductance Values

References

- [1] P. -H. Huang, C. -S. Chiu, G. -W. Huang, K. -M. Chen and L. -K. Wu, "A Low-Power Low-Noise W-band LNA in 90-nm CMOS Process With Source Degeneration Technique," in IEEE Microwave and Wireless Technology Letters, vol. 34, no. 1, pp. 69-71, Jan. 2024, doi: 10.1109/LMWT.2023.3331539. keywords: {Noise measurement;Gain measurement;Transmission line measurements;CMOS process;Semiconductor device measurement;Bandwidth;Wireless communication;Low-noise amplifiers;Millimeter wave communication;Low power;low-noise amplifier (LNA);millimeter-wave (MMW);source degeneration;W-band},
- [2] D. Sam Chrisvin, M. Dharshini, S. N. Senthilkumar and T. Jaspar Vinitha Sundari, "Design and Study of 90nm CMOS Common Source 2.4GHz Low Noise Amplifier," 2022 6th International Conference on Computing Methodologies and Communication (ICCMC), Erode, India, 2022, pp. 545-550, doi: 10.1109/ICCMC53470.2022.9753775. keywords: {Wireless communication;Noise figure;Low-noise amplifiers;Systematics;CMOS technology;Topology;Reflection coefficient;Low power applications;CMOS;90nm;Common Source Inductive load;Inductive Source Degeneration and Cascode Common Source LNA},
- [3] E. C. Becerra-Alvarez, J. M. de la Rosa and F. Sandoval, "Design of a 1-V 90-nm CMOS folded cascode LNA for multi-standard applications," 2010 53rd IEEE International Midwest Symposium on Circuits and Systems, Seattle, WA, USA, 2010, pp. 185-188, doi: 10.1109/MWSCAS.2010.5548704. keywords: {CMOS technology;Circuit optimization;Integrated circuit technology;Circuit noise;Low-noise amplifiers;Transceivers;Network topology;Circuit topology;Tuning;Frequency},
- [4] https://www.cambridge.org/us/files/8113/6698/2364/HFIC_chapter_7_low-noise_amplifier_design.pdf
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- [6] <http://smirc.stanford.edu/spiralCalc.html>